A

Study workflow

OpenPedCan v15 · pediatric CNS tumor biospecimens

Clinical and histologic annotation

Fusion annotation

TPM-normalized gene-expression matrix

Cohort assembly n = 343 pediatric-type diffuse HGG
one RNA-sequenced specimen per participant (245 initial · 40 recurrent/progressive · 58 autopsy)
DMG, H3 K27-altered 175 · pHGG, H3/IDH-wildtype 127 · DHG, H3 G34-mutant 31 · infant-type hemispheric 10

Immune-feature generation
CIBERSORTx LM22 (22 immune cell types) · orthogonal deconvolution (xCell, quanTIseq, MCP-counter, EPIC)
24 brain-tuned ssGSEA signatures → z-scored 46-feature immune matrix

Locked k = 3 assignment
500-replicate bootstrap consensus clustering (80% item resampling, average linkage); modal label over seeds 42–66
Cluster-number evaluation and resampling-stability analysis: 1,000 replicates, k = 2–10
k = 2 most stable; k = 3 retained as an exploratory three-state representation

Immune-high
n = 122

Immune-intermediate
n = 144

Immune-low
n = 77

Downstream analyses
integrated molecular group and anatomical location · sequential PERMANOVA · molecular-group-adjusted differential expression
Kaplan–Meier and multivariable Cox · single-cell and spatial support · sensitivity analyses

B

Analysis-set derivation

OpenPedCan v15
pediatric CNS tumor biospecimens

Excluded: repeat biospecimens, diagnoses outside the cohort, and 6 non-diffuse-HGG specimens (5 non-HGG; 1 methylation-class PXA)

Main cohort n = 343
one RNA-sequenced specimen per participant
DMG 175 · pHGG-WT 127 · DHG 31 · IHG 10

immune deconvolution, signature scoring, consensus clustering and molecular-group analyses used all 343 specimens

Anatomical-location analyses
n = 326
contingency tests and Cramér's V

− 17 other or unannotated sites

Survival analyses
n = 245 (206 deaths)
Kaplan–Meier and multivariable Cox

− 98 no usable OS record

Sequential PERMANOVA
n = 253
midline, hemispheric, posterior fossa

− 73 multi-compartment

Survival sensitivity
n = 187 delayed entry; n = 189 non-autopsy
alternative risk sets for the Immune-high association

Location and survival branches are not nested.